# Global Electric Vehicle Market Analysis

## Performance, Pricing, Battery Technology and Sales

SmartBridge Skill Wallet Project: Data Analytics with Tableau

Stack: MySQL, SQL, Tableau, Python, Flask

| | |
|---|---|
| Student name | _____ |
| Project link (GitHub) | _____ |
| Tableau Public link | _____ |
| Demo video link | _____ |

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# 1. Problem understanding

Electric vehicles are becoming a mainstream product, but manufacturers, policy makers and buyers still struggle to compare the market. Models vary widely in price, range, battery chemistry, charging speed and sales. Without a clear view of these factors it is hard to know which vehicle types sell best, what drives price, and which battery technologies last longest.

## Objectives

- Measure the size and structure of the EV market by category, manufacturer, region and country.
- Understand how price relates to range, battery capacity and motor power.
- Compare battery technologies (NMC, LFP, NCA, Sodium-Ion, Solid-State) on durability and charging.
- Identify high-value vehicles and the environmental impact ($CO_2$ saved).
- Deliver the findings as an interactive Tableau dashboard and story embedded in a Flask web app.

## Business questions

- Which categories and manufacturers lead in annual sales?
- How strongly do battery capacity and range drive price?
- Which battery type has the best cycle life and lowest degradation?
- How do market regions and manufacturing countries compare?
- Which vehicles give the most range per dollar?
- How much $CO_2$ do EVs save, by category?

# 2. Data collection and dataset description

The dataset **ev_market_test.csv** was provided by Skill Wallet. It has **100,000 rows and 37 columns**; each row is one EV model or variant (vehicle_id is unique and there are no duplicate records). Model years run from 2018 to 2026 with about 11,000 models per year.

| Group | Columns |
|---|---|
| Identity | vehicle_id, model_name, manufacturer (200), model_year, manufacturing_country (10), market_region (7), category (15) |
| Battery | battery_capacity_kwh, battery_type (5), battery_cycle_life, battery_degradation_pct_5yr |
| Performance | range_km, efficiency_wh_per_km, motor_power_kw, torque_nm, num_motors, drivetrain, top_speed_kmph, acceleration_0_100_sec |
| Charging | max_dc_charging_kw, charging_dc_fast_time_min, charging_ac_time_hr, num_charging_ports |
| Body and features | num_seats, weight_kg, cargo_capacity_liters, infotainment_screen_inch, autonomous_level, safety_rating_stars |
| Commercial | price_usd, insurance_cost_usd_yearly, warranty_years, resale_value_pct_3yr, annual_sales_units, customer_rating |
| Impact and use | co2_saved_tons_per_year, recommended_use_case (7) |

## Data quality findings

| Column | Missing values |
|---|---|
| resale_value_pct_3yr | 3,057 |
| battery_cycle_life | 2,028 |
| customer_rating | 1,931 |
| infotainment_screen_inch | 1,506 |
| annual_sales_units | 989 |

- All other columns are complete. Outliers exist by design (buses and trucks have very high efficiency_wh_per_km, seats up to 65, price up to about $410k), so charts are segmented by category.
- The data is synthetic: models per year, regional totals and average range by battery type are almost uniform. Visuals focus on fields with real variation (category, price, battery durability).

## 3. Storing data in the database

MySQL was used as the database. Create the database, then load the CSV with pandas and SQLAlchemy (script `scripts/load_to_db.py`).

```
CREATE DATABASE IF NOT EXISTS ev_db;

# scripts/load_to_db.py (core)
df = pd.read_csv("data/raw/ev_market_test.csv")
df.to_sql("ev_market", engine, if_exists="replace", index=False, chunksize=5000)
print(f"Loaded {len(df):,} rows into ev_market")
```

*Add screenshot here: MySQL Workbench showing the ev_market table and the row count (100,000).*

## 4. SQL operations

The queries below are saved in `sql/queries.sql`. They cover aggregation, grouping, ranking, calculated columns, a null audit and a view for Tableau.

```
-- Top 10 manufacturers by sales
SELECT manufacturer, SUM(annual_sales_units) AS total_sales
FROM ev_market GROUP BY manufacturer ORDER BY total_sales DESC LIMIT 10;

-- Battery technology comparison
SELECT battery_type, ROUND(AVG(range_km),1) AS avg_range_km,
       ROUND(AVG(charging_dc_fast_time_min),1) AS avg_fast_charge_min,
       ROUND(AVG(battery_cycle_life),0) AS avg_cycle_life,
       ROUND(AVG(battery_degradation_pct_5yr),2) AS avg_degradation_pct
FROM ev_market GROUP BY battery_type ORDER BY avg_cycle_life DESC;

-- Best value vehicles
SELECT model_name, manufacturer, category, price_usd, range_km,
       ROUND(range_km / price_usd * 1000, 2) AS km_per_1000usd
FROM ev_market WHERE price_usd > 0 ORDER BY km_per_1000usd DESC LIMIT 20;

-- Clean view used by Tableau
CREATE OR REPLACE VIEW vw_ev_clean AS
SELECT *, ROUND(range_km / price_usd * 1000, 2) AS range_per_1000usd,
  CASE WHEN price_usd < 20000 THEN 'Budget' WHEN price_usd < 60000 THEN 'Mid-range'
       WHEN price_usd < 120000 THEN 'Premium' ELSE 'Luxury' END AS price_segment,
  price_usd * annual_sales_units AS est_revenue_usd
FROM ev_market WHERE price_usd > 0 AND range_km > 0;
```

Other queries in the file: sales by category, region summary, year trend, manufacturing country summary, use-case summary and the missing-value audit. *Add screenshots of at least three query outputs here.*

## 5. Connecting the database to Tableau

- Install the MySQL ODBC/JDBC driver for Tableau Desktop.
- Connect > To a Server > MySQL; enter server, port 3306, username, password.
- Select database **ev_db** and drag the view **vw_ev_clean** to the canvas.
- Use a **Live** connection for testing and an **Extract** for the final workbook.

*Add screenshot here: Tableau data source page showing the connected view and 100,000 rows.*

# 6. Data preparation

- **Nulls:** left blank for averages (Tableau ignores nulls). Missing values are not imputed because they are under 3.1% per column.
- **Filtering:** the view removes rows with non-positive price or range (none found in this dataset).
- **Data types:** model_year kept as a discrete dimension; geographic role set on manufacturing_country.
- **High cardinality:** 200 manufacturers are handled with a Top N parameter.
- **Derived fields:** Price Segment, Range Class, Fast Charge Rating, Range per $1000 and Revenue (Est.).
- **Outliers:** buses, trucks and scooters are compared within their category rather than removed.

### Price segment distribution after preparation

| Segment | Rule | Models |
|---|---|---|
| Budget | price < $20,000 | 26,426 |
| Mid-range | $20,000 to $59,999 | 38,250 |
| Premium | $60,000 to $119,999 | 20,902 |
| Luxury | $120,000 and above | 14,422 |

# 7. Visualizations and calculated fields

The workbook contains 12 unique visualizations, 6 calculated fields and 6 filters.

| # | Visualization | Type | Question answered |
|---|---|---|---|
| 1 | Sales by category | Bar | Which vehicle types sell most? |
| 2 | Sales by model year | Line | Is the market growing? |
| 3 | Top N manufacturers | Bar + parameter | Who leads? |
| 4 | Manufacturing country | Filled map | Where are models built? |
| 5 | Region vs average price | Bar | How do regions compare? |
| 6 | Battery type share | Donut | Which chemistry dominates? |
| 7 | Cycle life and degradation | Bar / dual axis | Which battery lasts longest? |
| 8 | Price vs range | Scatter | Does more money buy more range? |
| 9 | Category x drivetrain | Heat map | Where do FWD/RWD/AWD concentrate? |
| 10 | Recommended use case | Treemap | What are EVs used for? |
| 11 | CO2 saved | Stacked bar | Where is the climate impact? |
| 12 | Customer rating | Histogram | How satisfied are owners? |

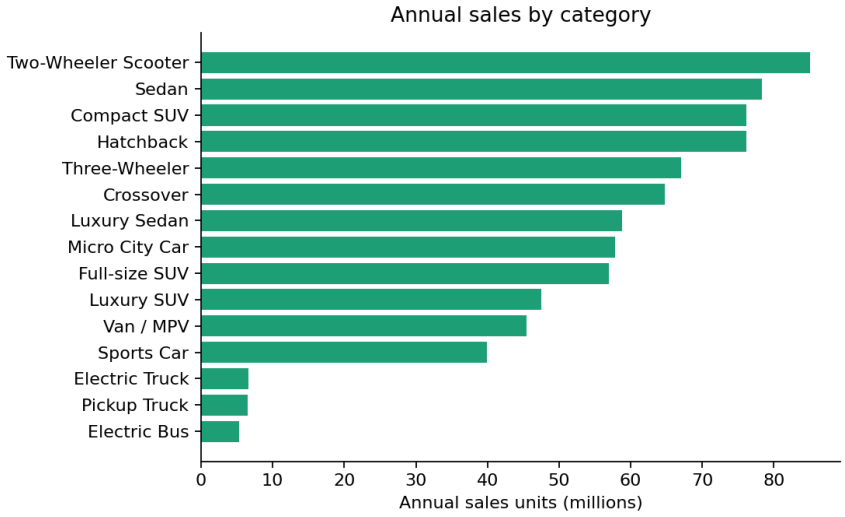
### Calculated fields

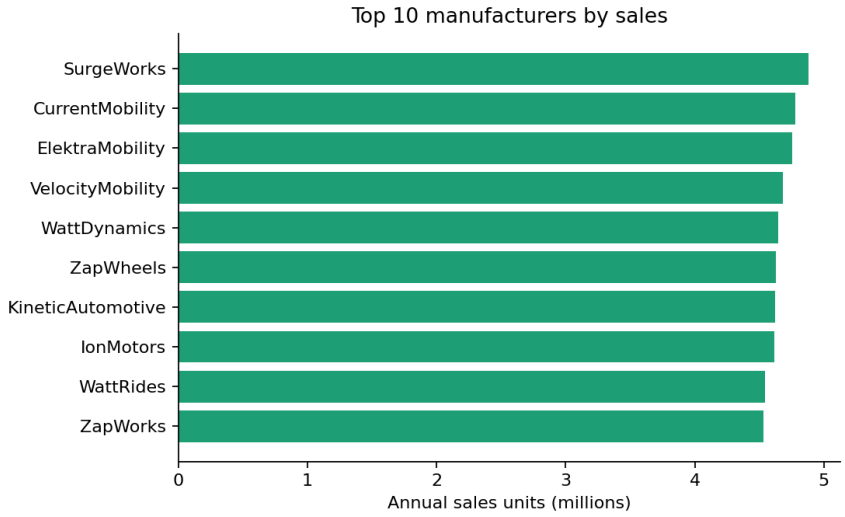
```
Range per $1000    = [range_km] / [price_usd] * 1000
Price Segment      = IF [price_usd] < 20000 THEN "Budget" ELSEIF [price_usd] < 60000 THEN "Mid-range"
                     ELSEIF [price_usd] < 120000 THEN "Premium" ELSE "Luxury" END
Range Class        = IF [range_km] >= 450 THEN "Long" ELSEIF [range_km] >= 250 THEN "Medium" ELSE "Short" END
Fast Charge Rating = IF [charging_dc_fast_time_min] <= 15 THEN "Ultra-fast"
                     ELSEIF [charging_dc_fast_time_min] <= 25 THEN "Fast" ELSE "Standard" END
Revenue (Est.)     = [price_usd] * [annual_sales_units]
Sales Share %      = SUM([annual_sales_units]) / TOTAL(SUM([annual_sales_units]))
```

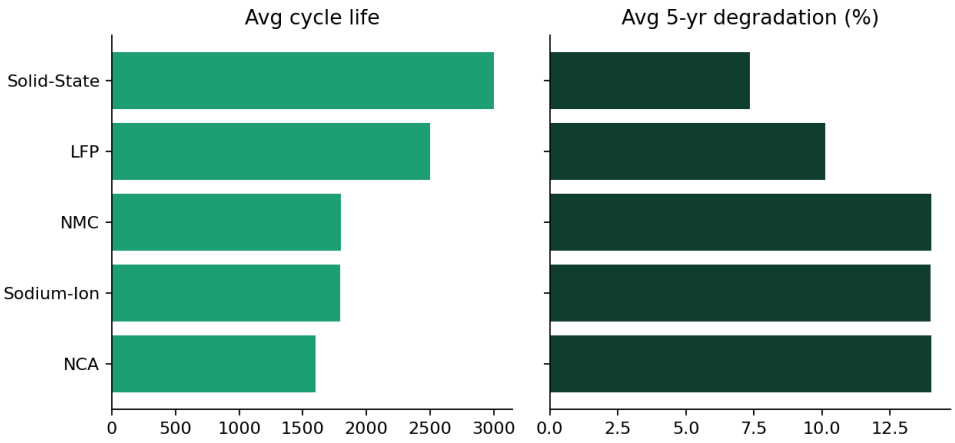
**Filters:** model_year, market_region, category, battery_type, Price Segment, Top N manufacturers (applied to all worksheets using the data source).

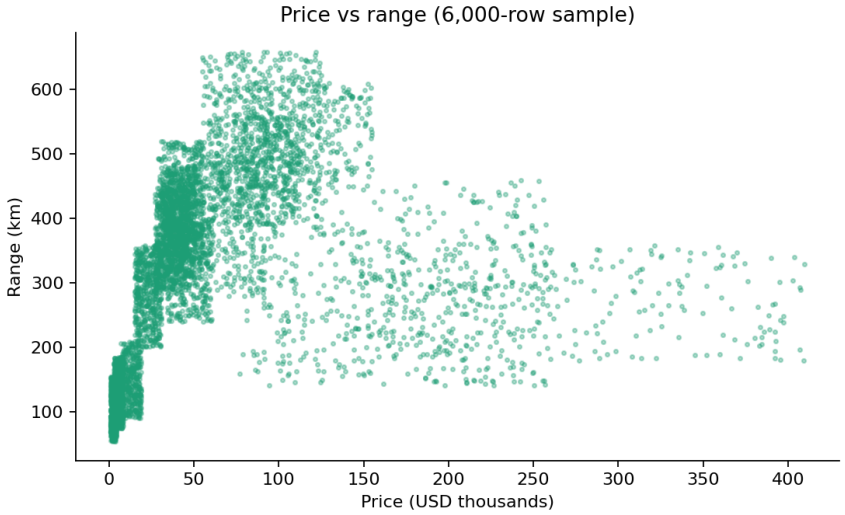
## Sample visual outputs (reproduced from the dataset)

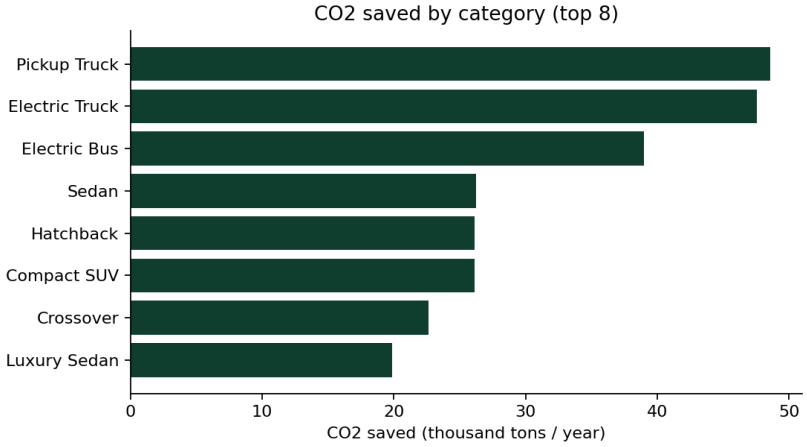
These charts were generated from the same data to show the expected patterns; rebuild them in Tableau and replace them with your own screenshots.




Avg cycle life
Avg 5-yr degradation (%)
Solid-State
LFP
NMC
Sodium-Ion
NCA
0
500
1000
1500
2000
2500
3000
0.0
2.5
5.0
7.5
10.0
12.5


Price vs range (6,000-row sample)
Range (km)
100
200
300
400
500
600
0
50
100
150
200
250
300
350
400
Price (USD thousands)


CO2 saved by category (top 8)
Pickup Truck
Electric Truck
Electric Bus
Sedan
Hatchback
Compact SUV
Crossover
Luxury Sedan
0
10
20
30
40
50
CO2 saved (thousand tons / year)

# 8. Dashboards

- **Market Overview:** KPI cards (total models, total sales, average price, average range) with sales by category, sales trend, top manufacturers and region vs price.
- **Technology:** battery share, cycle life vs degradation, price vs range scatter and the category x drivetrain heat map.
- **Geography and Impact:** manufacturing country map, use-case treemap, $CO_2$ saved and rating histogram.

## Responsive design

- Dashboard size set to **Automatic** so it fits the browser window.
- Phone and Tablet layouts created with Device Preview.
- Layout containers (horizontal/vertical) keep charts aligned; filters sit in a collapsible side panel.
- Dashboard actions: Use as Filter on category and region charts; Highlight on battery type.
- One green palette with consistent fonts and clear titles for readability.

*Add screenshots: desktop, tablet and phone layouts of each dashboard.*

# 9. Story (6 scenes)

| Scene | Title | Message |
|---|---|---|
| 1 | Market snapshot | 100,000 models, average price about $65,010, average range about 324 km. |
| 2 | Who sells the most | Two-wheeler scooters, sedans and compact SUVs lead; trucks, pickups and buses sell far fewer units. |
| 3 | Price vs range | Battery capacity drives price much more than range does. |
| 4 | Battery technology | Solid-State and LFP last longest with the lowest degradation. |
| 5 | Regions and countries | Regional totals are balanced; no single region or builder dominates. |
| 6 | Recommendations | Focus on high-volume categories and durable battery chemistries. |

## 10. Performance testing

| Parameter | Value |
|---|---|
| Rows rendered from DB | 100,000 |
| Columns | 37 (plus derived fields) |
| Data filters used | 6 |
| Calculated fields | 6 (plus Top N rank field) |
| Visualizations | 12 |
| Dashboards / story scenes | 3 / 6 |
| Live connection load time | _____ s (measure) |
| Extract load time | _____ s (measure) |

Measure load time with Help > Settings and Performance > Start Performance Recording, open each dashboard, then stop and screenshot the summary. Tips used to keep the dashboard fast: use an extract, hide unused fields, apply context filters, limit marks on scatter plots, and use the Top N parameter.

## 11. Web integration with Flask

Publish the workbook to Tableau Public and copy the view URLs into the `.env` file. The Flask app embeds them with the Tableau Embedding API.

```
# app.py (core)
@app.route("/dashboard")
def dashboard():
    return render_template("dashboard.html", viz_url=DASHBOARD_URL)

# templates/embed.html
<script type="module" src="https://public.tableau.com/javascripts/api/tableau.embedding.3.latest.min.js"></script>
<tableau-viz src="{{ viz_url }}" toolbar="bottom" hide-tabs></tableau-viz>
```

Routes: / (home), /dashboard, /story, /about. The CSS is responsive and supports light and dark mode. Run with `python app.py` and open http://127.0.0.1:5000.

*Add screenshots: home page, dashboard page and story page running in the browser.*

## 12. Key insights

- **Scooters lead volume:** Two-wheeler scooters sell about 85.0M units, followed by sedans (78.3M) and compact SUVs (76.1M). Electric trucks, pickups and buses sell only about 5 to 7M each.
- **Price follows battery size:** price correlates 0.84 with battery capacity but only 0.28 with range. Electric buses (average about $274k) and trucks (about $169k) are the most expensive categories.
- **Battery durability:** Solid-State averages about 2,999 charge cycles and 7.4% degradation over 5 years; LFP follows with about 2,498 cycles and 10.1%. NMC, NCA and Sodium-Ion are around 1,600 to 1,800 cycles and 14%.
- **No fast-charge or range gap:** every battery type averages about 21 minutes for DC fast charging and 320 to 325 km of range, so durability is the real differentiator.
- **Balanced market:** the top five manufacturers (SurgeWorks, CurrentMobility, ElektraMobility, VelocityMobility, WattDynamics) sell 4.6M to 4.9M units each, and regions and countries are evenly split.
- **Stable trend:** average price (about $65k) and range (about 324 km) are flat from 2018 to 2026, which suggests the data does not model technology improvement over time.

**Limitation:** because the data is synthetic, summed sales (about 773 million units) and estimated revenue are not real-world totals; treat them as relative measures.

# 13. Demo video plan, conclusion and future scope

## Demo video outline (about 8 to 10 minutes)

- 0:00 Problem statement and objectives
- 1:00 Dataset overview
- 2:00 MySQL table load and SQL queries
- 3:30 Tableau connection and data preparation
- 4:30 Visualizations and calculated fields
- 6:00 Dashboards and responsive layouts
- 7:00 Story
- 8:00 Flask app running with embedded dashboard and story
- 9:00 Insights, performance results and conclusion

## Conclusion

The project takes an EV market dataset from CSV to MySQL, analyses it with SQL, visualizes it in Tableau and publishes the dashboard and story in a Flask web app. The analysis shows which categories sell most, that battery size drives price, and that battery chemistry matters most for durability.

## Future scope

- Add real-world data (registrations, charging stations) to replace synthetic fields.
- Add forecasting of sales by category.
- Add user login and a live database refresh to the web app.
- Deploy the Flask app to a cloud platform.